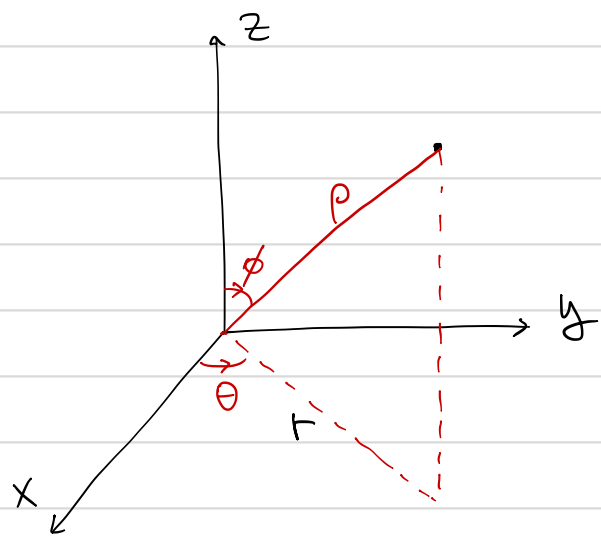
## Spherical coordinates (15.8)



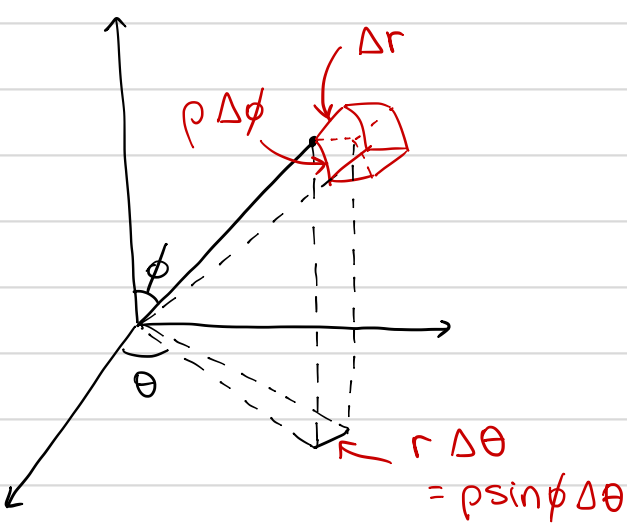

$z = \rho\cos\phi$ $\qquad$ $r = \rho\sin\phi$

$x = \rho\sin\phi\cos\theta$ $\qquad$ $\rho^2 = x^2+y^2+z^2$

$y = \rho\sin\phi\sin\theta$ $\qquad$ $= r^2+z^2$

$(\rho \geq 0,\ 0 \leq \theta < 2\pi,\ 0 \leq \phi < \pi)$

$$\boxed{dV = \rho^2\sin\phi\, d\rho\, d\phi\, d\theta}$$




**Example:** E = ball of radius a about the origin

$$\text{Vol} = \iiint_E 1\, dV$$

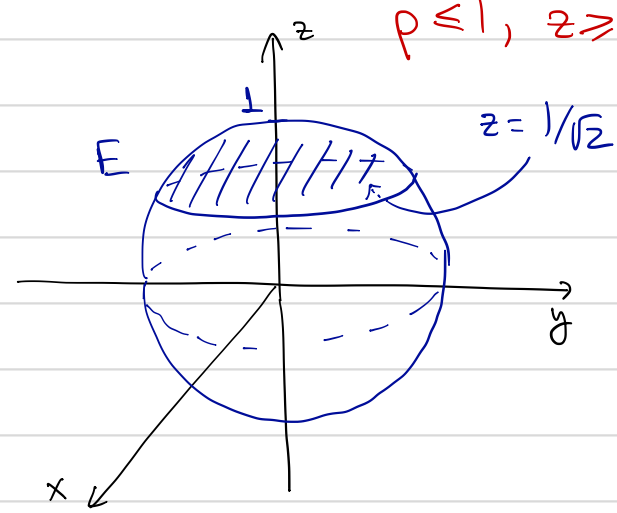
$$= \int_{\theta=0}^{2\pi}\int_{\phi=0}^{\pi}\int_{\rho=0}^{a} \rho^2\sin\phi\, d\rho\, d\phi\, d\theta$$

$$= 2\pi\cdot\left[-\cos\phi\right]_0^{\pi}\cdot\left[\frac{\rho^3}{3}\right]_0^{a}$$

$$= \frac{4\pi a^3}{3} \;//$$

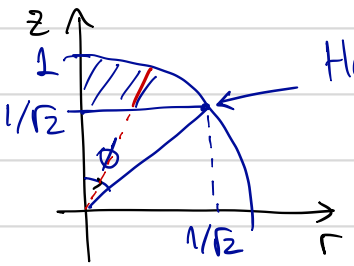
**Example:** Volume of cap

$\rho \leq 1,\ z \geq 1/\sqrt{2}$




$$\text{Vol} = \iiint_E dV =$$

$$= \int_{\theta=0}^{2\pi}\int_{\phi=0}^{\pi/4}\int_{\rho=\frac{1}{\sqrt{2}\cos\phi}}^{1} \rho^2\sin\phi\, d\rho\, d\phi\, d\theta$$




Here: $\rho = 1$ (sphere)

$z = 1/\sqrt{2}$

$\rho^2 = r^2 + z^2$

$\Rightarrow r^2 = 1 - 1/2$

$\Rightarrow r = 1/\sqrt{2}$

$\Rightarrow \phi = \pi/4$

$\rho\cos\phi = z$

$\Rightarrow \cos\phi = 1/\sqrt{2}$

$\Rightarrow \phi = \pi/4$

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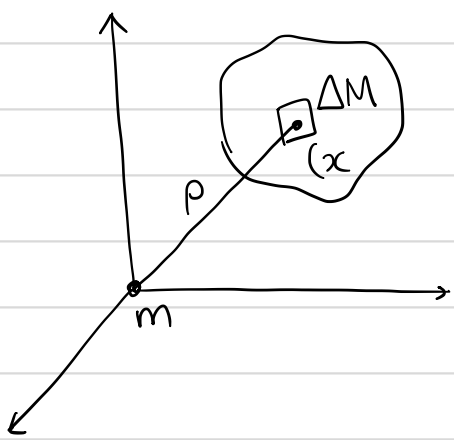
<u>Application</u>: Graviational attraction




$\Delta M = \delta\, dV$

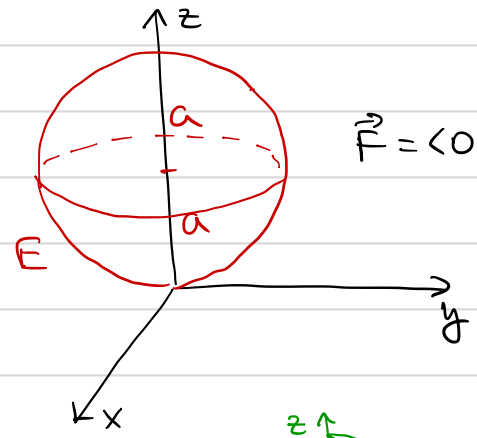
Magnitude: $\dfrac{G m \Delta M}{\rho^2}$

Direction: $\dfrac{\langle x, y, z\rangle}{\rho}$

$$\Rightarrow \vec{F} = \frac{G m \Delta M}{\rho^3}\langle x, y, z\rangle$$

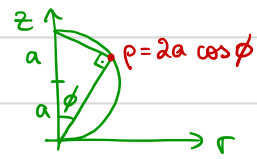
$$\vec{F}_{total} = G m \iiint_E \frac{\langle x, y, z\rangle}{\rho^3}\,\delta\, dV$$

<u>Example</u>: $\delta$ = constant.




$\vec{F} = \langle 0, 0, F\rangle$

$$\frac{F}{G m \delta} = \iiint_E \frac{z}{\rho^3}\, dV$$




$$= \int_{\theta=0}^{2\pi}\int_{\phi=0}^{\pi/2}\int_{\rho=0}^{2a\cos\phi} \frac{\cancel{\rho}\cos\phi}{\cancel{\rho^3}}\,\cancel{\rho^2}\sin\phi\, d\rho\, d\phi\, d\theta$$

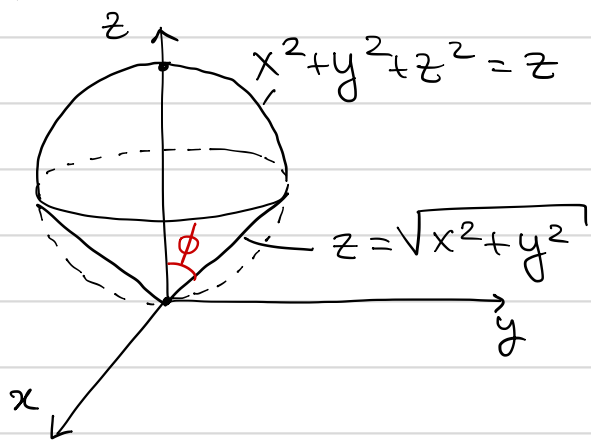
$$= 2\pi\int_0^{\pi/2} 2a\cos^2\phi\,\sin\phi\, d\phi$$

$$= 4\pi a\left[-\frac{\cos^3\phi}{3}\right]_0^{\pi/2} = \frac{4\pi a}{3} \;//$$

$$\Rightarrow F = G m \delta\,\frac{4\pi a}{3} = \frac{G m M}{a^2},$$

where $M = \dfrac{4\pi a^3}{3}\cdot\delta$ = mass(E).

<u>Example</u>: Volume of E:




$$x^2+y^2+\left(z-\tfrac{1}{2}\right)^2=1$$

sphere with center $(0,0,\frac{1}{2})$.

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In terms of $(\rho,\theta,\phi)$:

Sphere:

$$\rho^2=\rho\cos\phi \implies \rho=\cos\phi$$

Cone:

$$\rho\cos\phi=\rho\sin\phi$$
$$\implies \cos\phi=\sin\phi$$
$$\implies \phi=45^\circ=\pi/4.$$

$$\text{Vol}=\iiint_E dV$$

$$=\int_0^{\frac{\pi}{4}}\int_0^{2\pi}\int_0^{\cos\phi}\rho^2\sin\phi\,d\rho\,d\theta\,d\phi$$

$$=\int_0^{2\pi}d\theta\int_0^{\pi/4}\left[\frac{\rho^3}{3}\right]_0^{\cos\phi}\sin\phi\,d\phi$$

$$=\frac{2\pi}{3}\int_0^{\pi/4}\cos^3\phi\sin\phi\,d\phi$$

$$=\frac{2\pi}{3}\left[-\frac{\cos^4\phi}{4}\right]_0^{\pi/4}$$

$$=\frac{2\pi}{12}\left[-\left(\frac{\sqrt{2}}{2}\right)^4+1\right]=\frac{\pi}{8}\ //$$